

Technology Stack

Date	30 June 2026
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Jinja2	Used to build a responsive and interactive user interface. Jinja2 renders dynamic HTML pages with FastAPI.

2	Backend / Server-Side	Python, FastAPI, Uvicorn	FastAPI provides high-performance REST APIs, while Uvicorn efficiently serves the application for fast AI request processing.
3	Database / Data Storage	SQLite (or Local File Storage if you are not using a database)	Stores user information, learning history, quiz results, and application data efficiently. (If no database is implemented, write "Local File Storage" and mention uploaded files and logs are stored locally.)
4	Cloud / Hosting / Deployment	Not Used(Local File Storage)	The application temporarily stores uploaded files in the local uploads folder for AI Processing .A DataBase is not required for the current implimentation
5	Version Control & CI/CD	Git, GitHub	Used for source code management, version tracking, collaboration among team members, and maintaining project history.
6	Third-Party APIs / Other Tools	Google Gemini API, Transformers (LaMini-Flan-T5), Hugging Face, VS Code	Google Gemini powers AI-based educational responses, LaMini-Flan-T5 assists with NLP tasks, Hugging Face provides model support, and VS Code is used for development and debugging.